Tutorial - 10

Boosting

31/3/26

Sample	feature 1 (Age)	feature 2 (income)	feature 3 (credit score)	Target (loan Approved)
S_1	25	40	650	yes
S_2	30	60	700	yes
S_3	45	35	600	NO
S_4	50	30	580	NO
S_5	28	55	720	yes

Encoding

Target Encoding:

yes = +1

NO = -1

Questions

1. Assign initial weights to all 5 samples
2. construct a decision stump using feature 1 (Age) to classify the samples.
3. Identify misclassified samples and compute the weighted error rate.
4. calculate the classifier weight.
5. update the sample weights for the next iteration.

Sol:-

Desicion stump

↓ many

AdaBoost

1). Initial weights

$$w_i = \frac{1}{N} = \frac{1}{5} = 0.2$$

Target (y)	predict ($h_1(x_i)$)	w
Yes (+1) →	+1	0.2
Yes (+1) →	+1	0.2
NO (-1) →	-1	0.2
NO (-1) →	-1	0.2
Yes (+1) →	+1	0.2

2) Decision stump — decision boundary (taken)

Age $< 40 \rightarrow$ Yes (+1)

$\geq 40 \rightarrow$ NO (-1)

3) weighted Error

$$\boxed{\epsilon_t = \sum_{i=1}^{N} w_i \cdot I} \quad (y_i \neq h_t(x_i))$$

I → Indicator factor

$$\epsilon_1 = 0.2\times0 + 0.2\times0 + 0.2 + 0 + 0.2 + 0 + 0.2\times0]$$

$$\epsilon_1 = 0$$

4) classifier / model Error

$$\boxed{\alpha_t = \frac{1}{2} \ln\left(\frac{1-\epsilon_t}{\epsilon_t}\right)}$$

$$\alpha_1 = \frac{1}{2} \ln\left(\frac{1-0}{0}\right)$$

$$\alpha_1 = \infty$$

5) update weight

$$w_i^{new} \rightarrow w_i e^{-y_i \alpha_i h_i(x_i)}$$

$\downarrow$

$$\boxed{e^{-\infty} = 0}$$

if updated weight goes to '0', model will stop training

6) prediction

model will stop training

$$H(x) = h_1(x_i)$$

Target (y)	predicti ($h_1(x_i)$)	w_i	w_i^{new}	w^N
yes(+1)	+1	0.2	0.1	0.125
yes(+1)	+1	0.2	0.1	0.125
no(-1)	-1	0.2	0.1	0.125
no(-1)	-1	0.2	0.1	0.125
no(-1)	+1	0.2	0.4	0.5

3) weighted error:-

$$\varepsilon_1 = \sum w_i I \quad (y_i \neq h_i(x_i))$$

$$= 0.2 \times 1 = 0.2$$

4) classifier / model error :-

$$\alpha_1 = \frac{1}{2} \ln \left(\frac{1 - \epsilon_1}{\epsilon_1} \right)$$

$$= \frac{1}{2} \ln \left(\frac{1 - 0.2}{0.2} \right)$$

$$= 0.694$$

5) update weight :-

$$w_i^{new} \rightarrow w_i e^{-y_i \alpha_i h_i(x_i)}$$

$$0.2 \times e^{-[1 \times 0.694 \times 1]} = 0.1$$

for each sample update weight

6) normalized weight

$$\boxed{w_i^N = \frac{w_i}{\Sigma w_i}}$$

$$w_1^N = \frac{0.2}{0.1+0.1+0.1+0.1+0.4} = \frac{0.2}{0.8} = 0.125$$

$$w_5 = \frac{0.4}{0.8} = 0.5$$

7) prediction :

$$H(x) = \text{sign}\left[\sum_{t=1}^{T} \alpha_t h_t(x_i)\right]$$

$$= \text{sign}\left[\alpha_1 h_1(x_i) + \alpha_2 h_2(x_2) + \dots\right]$$

→ In exam they won't ask to calculate the complete prediction.